

Paper Title

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Abstract

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List of Corrections

UB :	Check above lemma.	6
UB :	Need to formalise this further.	6
RRS :	In the above expression, $\text{pay}_i(v_i, r_i)$ is the expected payment made by bidder i when his private valuation is v_i and his set of children is r_i . The expectation is taken over the types of the other bidders and is NOT conditioned on bidder i being reachable from the seller.	6
RRS :	This set needs to be defined formally	6
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RRS :	Add a crossrefs for the definitions and define the $g_i(\cdot)$ notation as some arguments are missing	9
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RRS :	Finish using concavity	13
SN :	we don't need both for all claims, but is it cleaner to use one consolidated definition rather than multiple ones? also, we can mention at the end of the proof that both conditions are not necessary for this result to hold. done! . .	14
SN :	done this far	17
UB :	Need to show that $\bar{z}$ exists, and that g is monotone, etc.	19
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1 Introduction

2 Preliminaries

Consider a directed graph $G = (N \cup \{s\}, E)$, where $N = \{1, \dots, n\}$ is the set of players involved in the auction of a single indivisible item and s is a distinguished node called the *seller*. The set E is the set of edges. Each edge (i, j) denotes that if node i shares information, node j will receive it. Typical examples of such graphs are online social networks where an individual can share information (selectively) with a subset of her neighbors. The direction signifies that almost all networks have asymmetric information flow (e.g., only followers receive the information from the followee).

In this network, node s is a single seller that wants to sell the indivisible item. Every other node $i \in N$ is a potential buyer and the information about the auction flows only via the direction of an edge. The information cannot reach a node unless there is a directed path from s to that node and each intermediate node decides to forward the information. An intermediate node may decide not to forward the information if it reduces its *utility*.

This setup naturally brings up an auction-like information-sharing game among the players. Each player $i \in N$ has a type $\theta_i = (v_i, r_i)$, where v_i is the valuation of agent i for the item, and r_i is the set of her directed neighbors. The set of valuations and subsets of neighbors of player i are denoted by $\mathcal{V}_i$ and $\mathcal{R}_i$ respectively, where $\mathcal{V}_i \subseteq [0, b_i]$. The type set of i , Θ_i , is therefore, $\mathcal{V}_i \times \mathcal{R}_i$, and agent i can report its type from this set. The information about the auction needs to reach via directed edges to player i for her to participate in the auction. Therefore, the auction asks every agent to report her valuation for the item and to forward the information to its directed neighbors. In our model, this is captured via their reported type $\hat{\theta}_i = (\hat{v}_i, \hat{r}_i)$ for every agent $i \in N$. We assume that the seller s is not a strategic player in this auction, rather he wants to sell the object and always forwards the information to his directed neighbors. The vector of the reported types of all the agents except i is denoted by $\hat{\theta}_{-i} = (\hat{\theta}_1, \dots, \hat{\theta}_{i-1}, \hat{\theta}_{i+1}, \dots, \hat{\theta}_n)$. We denote the set of all type profiles by $\Theta := \prod_{i \in N} \Theta_i$.

Depending on the reported types of the agents, particularly, the reported $\hat{r}_i$'s, the auction may reach only a subset of the agents in N . Throughout this paper, we will use the notation r_i to denote the *true* neighbor set of player i and assume that the reported $\hat{r}_i$ can only be a subset of it. To denote the reported valuation and directed neighbors on the subnetwork generated by $(\hat{\theta}_i, \hat{\theta}_{-i})$, we use a *filter function* fil for the graph G , and $\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i})$ denotes the *reduced type profile for the agents reachable from s when the agents reported the type profile $(\hat{\theta}_i, \hat{\theta}_{-i})$* . In this setup, the auction design goal is to incentivize each node i to truthfully reveal its private type θ_i . We characterize all such mechanisms in an elegant manner.

We consider auctions on this graph with randomized allocations to the agents. Formally, we define a *diffusion auction* in this setup as follows.

Definition 2.1 (Diffusion Auction). A diffusion auction (DA) is given by the *tuple (g, p) where g and p are the allocation and payment functions respectively. The allocation function*

$g : \Theta \rightarrow \Delta_n$ is such that its i -th component $g_i(\text{fil}(\theta))$ denotes the probability of agent i winning the object, where $\Delta_n := \{x \in \mathbb{R}_{\geq 0}^n : \sum_{i=1}^n x_i = 1\}$. Similarly, the payment function $p = (p_i)_{i \in N}$ is such that its i -th component $p_i : \Theta \rightarrow \mathbb{R}$ denotes the payment assigned to agent i .

Note that g_i should operate on the subnetwork that remains connected to s after the agents choose their actions $\hat{\theta}$. Hence the notation $g_i(\text{fil}(\cdot))$ is used in the definition above. Also, note that DA is different from the classical auction because the types of each agent now contain both the valuation (v_i) and the information on forwarding (r_i). The utility of agent i under DA is given by the standard quasi-linear model: $u_i^{(g,p)}((\hat{\theta}_i, \hat{\theta}_{-i}); \theta_i) = v_i g_i(\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i})) - p_i(\text{fil}(\hat{\theta}_i, \hat{\theta}_{-i}))$.

2.1 Useful facts

In this section we state a few facts that will be useful in later sections.

Fact 2.2. Consider a monotone non-decreasing function q and two functions λ and γ defined on the interval $x \in [a, b]$, such that $\exists x^*$ and

$$\lambda(x) \leq \gamma(x), \forall a \leq x \leq x^*, \quad \lambda(x) \geq \gamma(x), \forall x^* < x \leq b, \quad \text{and} \quad \int_a^b \lambda(x) dx = \int_a^b \gamma(x) dx,$$

i.e., functions λ and γ cross exactly at x^* in $[a, b]$ and they have the same integral, then the following must hold

$$\int_a^b q(x) \lambda(x) dx \geq \int_a^b q(x) \gamma(x) dx. \quad (1)$$

Proof. Consider the following expression

$$\begin{aligned} & \int_a^b q(x) (\lambda(x) - \gamma(x)) dx \\ &= \int_a^{x^*} q(x) (\lambda(x) - \gamma(x)) dx + \int_{x^*}^b q(x) (\lambda(x) - \gamma(x)) dx \end{aligned} \quad (2)$$

Note that the first expression on the RHS is at least $q(x^*) \int_a^{x^*} (\lambda(x) - \gamma(x)) dx$. This is because $\forall x \leq x^*$,

$$\begin{aligned} & q(x^*) \geq q(x) \\ \Rightarrow & q(x^*) (\lambda(x) - \gamma(x)) \leq q(x) (\lambda(x) - \gamma(x)). \end{aligned}$$

Using a similar argument for the second term of Equation (2), we get that the term is at least $q(x^*) \int_{x^*}^b (\lambda(x) - \gamma(x)) dx$. Combining these two inequalities, we complete Equation (2)

$$\int_a^b q(x) (\lambda(x) - \gamma(x)) dx$$

$$\begin{aligned}
&= \int_a^{x^*} q(x)(\lambda(x) - \gamma(x)) dx + \int_{x^*}^b q(x)(\lambda(x) - \gamma(x)) dx \\
&\geq q(x^*) \left(\int_a^{x^*} (\lambda(x) - \gamma(x)) dx + \int_{x^*}^b (\lambda(x) - \gamma(x)) dx \right) \\
&= q(x^*) \int_a^b (\lambda(x) - \gamma(x)) dx \\
&= 0.
\end{aligned}$$

This completes the proof. $\square$

3 Bayesian Setup and Design Desiderata

In this section, we consider auctions where the prior distribution over (θ_i, θ_{-i}) is given by P and is a common knowledge. In addition, we also assume that the distribution of valuations and neighbors are independent across agents and independent across these two parameters v_i, r_i for every agent. Formally, $P(\theta_i, \theta_{-i}) = \prod_{i \in N} f_i(v_i) h_i(r_i)$, where f_i and h_i are the distributions of the valuations v_i and the neighbors r_i , for each $i \in N$. First, we define the notion of truthfulness in the prior-based setup.

Definition 3.1 (Diffusion Bayesian Incentive Compatibility). *A DA (g, p) on a graph G is diffusion Bayesian incentive compatible (DBIC) if every agent's expected utility is maximized by reporting her true type irrespective of what the other agents report, i.e., for every $i \in N$, $\forall \theta_i = (v_i, r_i)$,*

$$\begin{aligned}
&\mathbb{E}[v_i g_i(\text{fil}((v_i, r_i), (v_{-i}, r_{-i}))) - p_i(\text{fil}((v_i, r_i), (v_{-i}, r_{-i}))) \mid \theta_i = (v_i, r_i)] \\
&\geq \mathbb{E}[v_i g_i(\text{fil}((v'_i, r'_i), (v_{-i}, r_{-i}))) - p_i(\text{fil}((v'_i, r'_i), (v_{-i}, r_{-i}))) \mid \theta_i = (v_i, r_i)], \forall v'_i, r'_i \subseteq r_i.
\end{aligned}$$

For convenience, we define the *expected* allocation and payments with the shorthand notation as described below.

$$\alpha_i(v_i, r_i) = \mathbb{E}[g_i(\text{fil}(\theta_i, \theta_{-i})) \mid \theta_i = (v_i, r_i)] \quad \textbf{(allocation)} \quad (3)$$

$$\text{pay}_i(v_i, r_i) = \mathbb{E}[p_i(\text{fil}(\theta_i, \theta_{-i})) \mid \theta_i = (v_i, r_i)] \quad \textbf{(payment)} \quad (4)$$

Since we assume that the agent types are independent, in this case we do not need to condition on θ_i in the above definitions.

In the Bayesian setup, the notion of participation guarantee is given by *interim individual rationality (IIR)* where the expected utility of a player to join the mechanism is non-negative after she learns her own type.

Definition 3.2 (Interim Individual Rationality). *A DA (g, p) on a graph G is interim individually rational (IIR) if $v_i \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq 0$, $\forall v_i, r_i, \forall i \in N$.*

Note that a necessary condition for IIR is that $\text{pay}_i(0, r_i) \leq 0, \forall i \in N$.

3.1 Characterization of DBIC Mechanisms

Our first result is to characterize the DBIC auctions.

Definition 3.3 (Monotone and Forwarding-Friendliness in Expectation (MFFE)). *For a given network G , a DA (g, p) is monotone and forwarding-friendly in expectation (MFFE) if, for every agent $i \in N$:*

1. *For every $r_i \in \mathcal{R}_i$, the expected allocation function $\alpha_i(v_i, r'_i)$ is non-decreasing in v_i .*
2. *For every $(v_i, r_i) \in \mathcal{V}_i \times \mathcal{R}_i$ and for every $r'_i \subseteq r_i$, the following payment formula is satisfied.*

$$\text{pay}_i(v_i, r'_i) = \text{pay}_i(0, r'_i) + v_i \alpha_i(v_i, r'_i) - \int_0^{v_i} \alpha_i(y, r'_i) dy \quad (5)$$

3. *For every $(v_i, r_i) \in \mathcal{V}_i \times \mathcal{R}_i$ and every $r'_i \subseteq r_i \in \mathcal{R}_i$,*

$$v_i \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \alpha_i(v_i, r'_i) - \text{pay}_i(v_i, r'_i). \quad (6)$$

Or, substituting for $\text{pay}_i(v_i, r_i)$ and $\text{pay}_i(v_i, r'_i)$ from Equation (5), the values of $\text{pay}_i(0, r'_i)$ and $\text{pay}_i(0, r_i)$ must satisfy the following inequality .

$$\text{pay}_i(0, r'_i) - \text{pay}_i(0, r_i) \geq \int_0^{v_i} (\alpha_i(y, r'_i) - \alpha_i(y, r_i)) dy \quad (7)$$

We can now state the following characterization result. The proof is deferred to the appendix.

Theorem 3.4 (DBIC Characterization). *A DA (g, p) is DBIC if and only if it is MFFE.*

Also note that, if we need (g, p) to be both DBIC and IIR, it reduces to $\text{pay}_i(0, r_i) \leq 0$.

Lemma 3.5. *Let (g, p) be DBIC and IIR. Then there is another DA (g, p') that is DBIC and IIR, and satisfies (i) $\text{pay}'_i(0, \emptyset) = 0$ for all agents i , and (ii) $p'_i(\text{fil}(\theta)) \geq p_i(\text{fil}(\theta))$ for all agents i and types θ .*

Proof. Since (g, p) is IIR, we know that $\text{pay}_i(0, \emptyset) \leq 0$ for all $i \in N$. For each agent $i \in N$, we define $p'_i(\theta) = p_i(\theta) - \text{pay}_i(0, \emptyset)$. The payment function p' clearly satisfies the two conditions in the lemma statement. It remains to show that (g, p') is DBIC and IIR.

To check that the modified mechanism is DBIC, since the same quantity is subtracted from each payment, the conditions in Equation (5) and Equation (7) are maintained. To check IIR, in Equation (7), the right hand side is zero when $v_i = 0$, and hence $\text{pay}_i(0, r'_i) \geq$

$\text{pay}_i(0, r_i)$ for $r'_i \subseteq r_i$. In particular, for any set r_i , $\text{pay}_i(0, \emptyset) \geq \text{pay}_i(0, r_i)$. Hence subtracting $\text{pay}_i(0, \emptyset)$ from each payment maintains that if agents report 0 value, their expected payment is nonpositive, and hence the modified mechanism is IIR. $\square$

UB: Check above lemma.

UB!

In the following, our goal is to maximize revenue. We can hence assume that $\text{pay}_i(0, \emptyset) = 0$ for all agents.

Realizable expected allocations. We say that expected allocation function $(\alpha_i)_{i \in N}$ are *realizable* if there exists a DBIC and IIR DA (g, p) for which $\alpha_i(v_i, r_i) = \mathbb{E}[g_i(\text{fil}(\theta_i, \theta_{-i})) \mid \theta_i = (v_i, r_i)]$. For this it is necessary that the α_i 's must be monotone, but this is not sufficient. Another necessary condition is that $\sum_{i \in N} \mathbb{E}_{(v_i, r_i)} \alpha_i(v_i, r_i) \leq 1$. We denote by $\mathcal{F}$ the set of *realizable* expected allocations. **UB:** Need to formalise this further.

UB!

4 The Revenue Maximization Problem

From the setup described in the previous section, it is easy to see that the expected revenue of the seller is given by:

$$\sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \text{pay}_i(v_i, r_i) dv_i.$$

RRS: In the above expression, $\text{pay}_i(v_i, r_i)$ is the expected payment made by bidder i when his private valuation is v_i and his set of children is r_i . The expectation is taken over the types of the other bidders and is NOT conditioned on bidder i being reachable from the seller. Our goal is to optimize the revenue subject to the BIC (that are equivalent to the conditions in Definition 3.3, by Theorem 3.4), IIR constraints, and that the α_i 's can be realized. Let $\mathcal{F}$ denote the set of realizable α_i 's.¹ **RRS:** This set needs to be defined formally. This means that we have to solve the problem:

RRS!

$$\begin{aligned} \max \quad & \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \text{pay}_i(v_i, r_i) dv_i \\ \text{s.t.} \quad & v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v'_i, r'_i) - \text{pay}_i(v'_i, r'_i), \quad \forall i, (v_i, r_i), (v'_i, r'_i) : r'_i \subseteq r_i \\ & \text{pay}_i(0, r_i) \leq 0, \quad \forall i, r_i \\ & (\alpha_i)_{i \in N} \in \mathcal{F}. \end{aligned}$$

RRS!

¹Note that this also includes Item 1 of Definition 3.3.

RRS!

RRS: By which lemma?, we observe that these constraints are equivalent to:

$$\begin{aligned}
\max \quad & \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \text{pay}_i(v_i, r_i) dv_i \\
\text{s.t.} \quad & v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v'_i, r_i) - \text{pay}_i(v'_i, r_i), \quad \forall i, (v_i, r_i), v'_i \\
& v_i \cdot \alpha_i(v_i, r_i) - \text{pay}_i(v_i, r_i) \geq v_i \cdot \alpha_i(v_i, r'_i) - \text{pay}_i(v_i, r'_i), \quad \forall i, (v_i, r_i), r'_i \subseteq r_i \\
& \text{pay}_i(0, r_i) \leq 0, \quad \forall i, r_i \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

Applying the Myerson transformation, we get that the first constraint is equivalent to saying that $\alpha_i(\cdot, r_i)$ is monotone non-decreasing and we have $\text{pay}_i(v_i, r_i) = \text{pay}_i(0, r_i) + \int_0^{v_i} x d\alpha_i(x, r_i) = \text{pay}_i(0, r_i) + v_i \cdot \alpha_i(v_i, r_i) - \int_0^{v_i} \alpha_i(x, r_i) dx$. **RRS**: As monotonicity is implied by virtue of the last constraint, and using the shorthands $\text{pay}_i(r_i) = \text{pay}_i(0, r_i)$ and $\varphi_i(v_i) = v_i - \frac{1-F_i(v_i)}{f_i(v_i)}$, we get that this is equivalent to:

$$\begin{aligned}
\max \quad & \sum_i \sum_{r_i} h_i(r_i) \text{pay}_i(r_i) + \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \varphi_i(v_i) \alpha_i(v_i, r_i) dv_i \\
\text{s.t.} \quad & \text{pay}_i(r'_i) - \text{pay}_i(r_i) \geq \int_0^{v_i} \alpha_i(x, r'_i) dx - \int_0^{v_i} \alpha_i(x, r_i) dx, \quad \forall i, (v_i, r_i), r'_i \subseteq r_i \\
& \text{pay}_i(r_i) \leq 0, \quad \forall i, r_i \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

4.1 Restricting to at most One Forward Node

We now specialize the above optimization problem to the case when every node has at most one other node it can forward to, or equivalently, to the case when $\mathcal{R}_i$ is either empty or singleton for all i . Let $E \subseteq N$ be the set of all i for which $\mathcal{R}_i = \emptyset$. The above program is equivalent to:

$$\begin{aligned}
\max \quad & \sum_i \sum_{r_i} h_i(r_i) \text{pay}_i(r_i) + \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \varphi_i(v_i) \alpha_i(v_i, r_i) dv_i \\
\text{s.t.} \quad & \text{pay}_i(\emptyset) - \text{pay}_i(\mathcal{R}_i) \geq \int_0^{v_i} \alpha_i(x, \emptyset) dx - \int_0^{v_i} \alpha_i(x, \mathcal{R}_i) dx, \quad \forall i, v_i : i \notin E \quad (8) \\
& \text{pay}_i(r_i) \leq 0, \quad \forall i, r_i \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

Next, we consider a relaxed version of the above optimization problem that relaxes the first constraint and only requires it when $v_i = b_i$ and when $v_i = 0$. As ?? shows that **RRS**: state and prove this lemma there always exists an optimal solution for the relaxed program that satisfies all the constraints in the original program, we have that the optimal value

of the two programs are the same. Thus, we can focus on the relaxed program (where we change the variable x to v_i) **RRS**: verify this

RRS!

$$\begin{aligned}
\max \quad & \sum_i \sum_{r_i} h_i(r_i) \text{pay}_i(r_i) + \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \varphi_i(v_i) \alpha_i(v_i, r_i) dv_i \\
\text{s.t.} \quad & \text{pay}_i(\emptyset) - \text{pay}_i(\mathcal{R}_i) \geq \max\left(0, \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i\right), \quad \forall i \notin E \quad (9) \\
& \text{pay}_i(r_i) \leq 0, \quad \forall i, r_i \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

Next, we show ?? that shows that **RRS**: state and prove this lemma there always exists an optimal solution for the above optimization problem that additionally satisfies that the maximum is not attained at 0. Thus, we can add that as a constraint and get the following equivalent formulation.

RRS!

$$\begin{aligned}
\max \quad & \sum_i \sum_{r_i} h_i(r_i) \text{pay}_i(r_i) + \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \varphi_i(v_i) \alpha_i(v_i, r_i) dv_i \\
\text{s.t.} \quad & \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i \geq 0, \quad \forall i \notin E \\
& \text{pay}_i(\emptyset) - \text{pay}_i(\mathcal{R}_i) \geq \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i, \quad \forall i \notin E \quad (10) \\
& \text{pay}_i(r_i) \leq 0, \quad \forall i, r_i \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

Observe that the constraints above allow us to solve for the value of $\text{pay}_i(r_i)$ in the optimal solution. The value of $\text{pay}_i(\emptyset)$ in the optimal solution is 0 for all i and the value of $\text{pay}_i(\mathcal{R}_i)$ is $-\left(\int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i\right)$ for all $i \in E$. Plugging in, we get:

$$\begin{aligned}
\max \quad & \sum_{i \notin E} h_i(\mathcal{R}_i) \cdot \left(\int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i - \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i \right) \\
& + \sum_i \sum_{r_i} \int_0^{b_i} f_i(v_i) h_i(r_i) \varphi_i(v_i) \alpha_i(v_i, r_i) dv_i \\
\text{s.t.} \quad & \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i \geq 0, \quad \forall i \notin E \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

Recall that E is the set of all i satisfying $\mathcal{R}_i = \emptyset$. The objective function is the same as:

$$\begin{aligned}
\max \quad & \sum_i \int_0^{b_i} h_i(\emptyset) \alpha_i(v_i, \emptyset) \cdot \left(f_i(v_i) \varphi_i(v_i) - \frac{h_i(\mathcal{R}_i) \cdot \mathbf{1}(i \notin E)}{h_i(\emptyset)} \right) dv_i \\
& + \sum_{i \notin E} \int_0^{b_i} h_i(\mathcal{R}_i) \alpha_i(v_i, \mathcal{R}_i) \cdot (f_i(v_i) \varphi_i(v_i) + 1) dv_i \\
\text{s.t.} \quad & \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i \geq 0, \quad \forall i \notin E \\
& (\alpha_i)_{i \in N} \in \mathcal{F}.
\end{aligned}$$

4.2 Specializing to a Chain of Buyers

Henceforth, we assume that there are at most n buyers and for all $0 < i < n$, we have $\mathcal{R}_i = \{i + 1\}$ meaning that buyer i can only forward to buyer $i + 1$. We also assume that $\mathcal{R}_n = \emptyset$ meaning that buyer n cannot forward to anyone. This implies that $E = \{n\}$. Using the notation $h_i^0 = h_i(\emptyset)$ for all $i \in [n]$ and $h_i^{+1} = h_i(\{i + 1\}) = h_i(\mathcal{R}_i)$ for all $0 < i < n$, we can rewrite the above program as:

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_0^{b_i} h_i^0 \cdot \alpha_i(v_i, \emptyset) \cdot \left(f_i(v_i) \varphi_i(v_i) - \frac{h_i^{+1} \cdot \mathbf{1}(i < n)}{h_i^0} \right) dv_i \\
& + \sum_{i=1}^{n-1} \int_0^{b_i} h_i^{+1} \cdot \alpha_i(v_i, \mathcal{R}_i) \cdot (f_i(v_i) \varphi_i(v_i) + 1) dv_i \\
\text{s.t.} \quad & \int_0^{b_i} \alpha_i(v_i, \emptyset) dv_i - \int_0^{b_i} \alpha_i(v_i, \mathcal{R}_i) dv_i \geq 0, \quad \forall i < n \\
& (\alpha_i)_{i \in [n]} \in \mathcal{F}.
\end{aligned}$$

We now plug in the definitions of $\mathcal{F}$, $\alpha_i(v_i, \emptyset)$ for $i \in [n]$ and $\alpha_i(v_i, \mathcal{R}_i)$ for $0 < i < n$ **RRS**: Add a crossrefs for the definitions and define the $g_i(\cdot)$ notation as some arguments are missing into the above program. In this case, using the notation $H_i = h_i^0 \cdot \prod_{i'=1}^{i-1} h_{i'}^{+1}$ for all $i \in [n]$, the definitions for α_i simplify to: **RRS**!

$$\begin{aligned}
\alpha_i(v_i, \emptyset) &= \frac{H_i}{h_i^0} \cdot \int_{v_{<i}} f_{<i}(v_{<i}) \cdot g_i(v_{\leq i}) dv_{<i}, \\
\alpha_i(v_i, \mathcal{R}_i) &= \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot \int_{v_{[i'] \setminus \{i\}}} f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) dv_{[i'] \setminus \{i\}}.
\end{aligned}$$

Plugging these in the optimization problem above, we get: **RRS**: Write $\mathcal{V}$ in the limit of **RRS**!

the integral.

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} H_i \cdot f_{<i}(v_{<i}) \cdot g_i(v_{\leq i}) \cdot \left(f_i(v_i) \varphi_i(v_i) - \frac{h_i^{+1} \cdot \mathbb{1}(i < n)}{h_i^0} \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} H_{i'} \cdot f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) \cdot (f_i(v_i) \varphi_i(v_i) + 1) dv_{\leq i'} \\
\text{s.t.} \quad & \frac{H_i}{h_i^0} \cdot \int_{v_{\leq i}} f_{<i}(v_{<i}) \cdot g_i(v_{\leq i}) dv_{\leq i} \geq \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot \int_{v_{\leq i'}} f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) dv_{\leq i'}, \quad \forall i < n \\
& \sum_{i=1}^{i'} g_i(v_{\leq i'}) \leq 1, \quad \forall i' \in [n], v_{\leq i'} \\
& g_i(v_{\leq i'}) \geq 0, \quad \forall i \leq i' \in [n], v_{\leq i'}.
\end{aligned}$$

To continue, for all $i \leq i' \in [n]$, we define an extra variable $Z_{i,i'} = \int_{v_{\leq i'}} f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) dv_{\leq i'}$. As these variables are fresh, we can add them to the constraints without changing the optimization problem. We get:

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} H_i \cdot f_{<i}(v_{<i}) \cdot g_i(v_{\leq i}) \cdot \left(f_i(v_i) \varphi_i(v_i) - \frac{h_i^{+1} \cdot \mathbb{1}(i < n)}{h_i^0} \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} H_{i'} \cdot f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) \cdot (f_i(v_i) \varphi_i(v_i) + 1) dv_{\leq i'} \\
\text{s.t.} \quad & \frac{H_i}{h_i^0} \cdot \int_{v_{\leq i}} f_{<i}(v_{<i}) \cdot g_i(v_{\leq i}) dv_{\leq i} \geq \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot \int_{v_{\leq i'}} f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) dv_{\leq i'}, \quad \forall i < n \\
& \sum_{i=1}^{i'} g_i(v_{\leq i'}) \leq 1, \quad \forall i' \in [n], v_{\leq i'} \\
& Z_{i,i'} = \int_{v_{\leq i'}} f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) dv_{\leq i'}, \quad \forall i \leq i' \in [n] \\
& g_i(v_{\leq i'}) \geq 0, \quad \forall i \leq i' \in [n], v_{\leq i'}.
\end{aligned}$$

This is equivalent to:

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} H_i \cdot f_{<i}(v_{<i}) \cdot g_i(v_{\leq i}) \cdot \left(f_i(v_i) \varphi_i(v_i) - \frac{h_i^{+1} \cdot \mathbb{1}(i < n)}{h_i^0} \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} H_{i'} \cdot f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) \cdot (f_i(v_i) \varphi_i(v_i) + 1) dv_{\leq i'} \\
\text{s.t.} \quad & \frac{H_i}{h_i^0} \cdot Z_{i,i} \geq \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot Z_{i,i'}, \quad \forall i < n
\end{aligned}$$

$$\begin{aligned}
\sum_{i=1}^{i'} g_i(v_{\leq i'}) &\leq 1, & \forall i' \in [n], v_{\leq i'} \\
Z_{i,i'} &= \int_{v_{\leq i'}} f_{[i'] \setminus \{i\}}(v_{[i'] \setminus \{i\}}) \cdot g_i(v_{\leq i'}) dv_{\leq i'}, & \forall i \leq i' \in [n] \\
g_i(v_{\leq i'}) &\geq 0, & \forall i \leq i' \in [n], v_{\leq i'}.
\end{aligned}$$

4.3 Specializing to Uniform Buyers

We now use the fact that the distribution for buyer i for all $i \in [n]$ is just the uniform distribution over the interval $[0, b_i]$ (for some $b_i > 0$). For all sets $I \subseteq [n]$, we define the notation $b_I = \prod_{i \in I} b_i$. Using this notation, we get:

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} \frac{H_i}{b_{<i}} \cdot g_i(v_{\leq i}) \cdot \left(\frac{2v_i}{b_i} - 1 - \frac{h_i^{+1} \cdot \mathbb{1}(i < n)}{h_i^0} \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} \frac{H_{i'}}{b_{[i'] \setminus \{i\}}} \cdot g_i(v_{\leq i'}) \cdot \frac{2v_i}{b_i} dv_{\leq i'} \\
\text{s.t.} \quad & \frac{H_i}{h_i^0} \cdot Z_{i,i} \geq \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot Z_{i,i'}, & \forall i < n \\
& \sum_{i=1}^{i'} g_i(v_{\leq i'}) \leq 1, & \forall i' \in [n], v_{\leq i'} \\
& Z_{i,i'} = \frac{1}{b_{[i'] \setminus \{i\}}} \cdot \int_{v_{\leq i'}} g_i(v_{\leq i'}) dv_{\leq i'}, & \forall i \leq i' \in [n] \\
& g_i(v_{\leq i'}) \geq 0, & \forall i \leq i' \in [n], v_{\leq i'}.
\end{aligned}$$

Note from the above that $\sum_{i=1}^{i'} \frac{Z_{i,i'}}{b_i} = \frac{1}{b_{\leq i'}} \cdot \sum_{i=1}^{i'} \int_{v_{\leq i'}} g_i(v_{\leq i'}) dv_{\leq i'} \leq \frac{1}{b_{\leq i'}} \cdot \int_{v_{\leq i'}} 1 \cdot dv_{\leq i'} = 1$. Next, we show ?? that shows that **RRS**: state and prove this lemma there always exists an optimal solution for the above optimization problem that additionally satisfies that there are (positive or negative) constants $(\Delta_{i,i'})_{i \leq i' \in [n]}$ such that for all $i \leq i' \in [n]$, we have $g_i(v_{\leq i'}) = 1$ if and only if $i = \arg \max_{0 \leq i'' \leq i'} \left(\frac{v_{i''}}{b_{i''}} + \Delta_{i'',i'} \right)$ (ties in the argmax broken arbitrarily) and 0 otherwise. Here and henceforth, we adopt the convention that $v_0 = b_0 = \frac{v_0}{b_0} = \Delta_{0,i'} = 0$. This means that we can restrict attention to the functions $g_i(\cdot)$ of this form. We get:

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} \frac{H_i}{b_{<i}} \cdot g_i(v_{\leq i}) \cdot \left(\frac{2v_i}{b_i} - 1 - \frac{h_i^{+1} \cdot \mathbb{1}(i < n)}{h_i^0} \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} \frac{H_{i'}}{b_{[i'] \setminus \{i\}}} \cdot g_i(v_{\leq i'}) \cdot \frac{2v_i}{b_i} dv_{\leq i'}
\end{aligned}$$

$$\begin{aligned}
\text{s.t. } \quad & \frac{H_i}{h_i^0} \cdot Z_{i,i} \geq \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot Z_{i,i'}, & \forall i < n \\
& \sum_{i=1}^{i'} g_i(v_{\leq i'}) \leq 1, & \forall i' \in [n], v_{\leq i'} \\
& Z_{i,i'} = \frac{1}{b_{[i'] \setminus \{i\}}} \cdot \int_{v_{\leq i'}} g_i(v_{\leq i'}) \, dv_{\leq i'}, & \forall i \leq i' \in [n] \\
& g_i(v_{\leq i'}) = \mathbb{1} \left(i = \arg \max_{0 \leq i'' \leq i'} \left(\frac{v_{i''}}{b_{i''}} + \Delta_{i'',i'} \right) \right), & \forall i \leq i' \in [n], v_{\leq i'} \\
& g_i(v_{\leq i'}) \geq 0, & \forall i \leq i' \in [n], v_{\leq i'}.
\end{aligned}$$

Removing the constraints that are implied and simplifying, we get:

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} \frac{H_i}{b_{< i}} \cdot g_i(v_{\leq i}) \cdot \left(\frac{2v_i}{b_i} - 1 - \frac{h_i^{+1} \cdot \mathbb{1}(i < n)}{h_i^0} \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} \frac{H_{i'}}{b_{[i'] \setminus \{i\}}} \cdot g_i(v_{\leq i'}) \cdot \frac{2v_i}{b_i} dv_{\leq i'} \\
\text{s.t. } \quad & \frac{H_i}{h_i^0} \cdot Z_{i,i} \geq \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot Z_{i,i'}, & \forall i < n \\
& Z_{i,i'} = \frac{1}{b_{[i'] \setminus \{i\}}} \cdot \int_{v_{\leq i'}} g_i(v_{\leq i'}) \, dv_{\leq i'}, & \forall i \leq i' \in [n] \\
& g_i(v_{\leq i'}) = \mathbb{1} \left(i = \arg \max_{0 \leq i'' \leq i'} \left(\frac{v_{i''}}{b_{i''}} + \Delta_{i'',i'} \right) \right), & \forall i \leq i' \in [n], v_{\leq i'}.
\end{aligned}$$

4.4 An Inductive Solution

To continue, we consider a more general optimization problem with additional parameters $(\Theta_i)_{i \in [n]}$ and $(\Phi_i)_{i \in [n]}$ that are non-negative. We emphasize that these are parameters and not variables in the program.

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \int_{v_{\leq i}} \frac{H_i}{b_{< i}} \cdot g_i(v_{\leq i}) \cdot \left(\frac{2v_i}{b_i} - 1 - \Phi_i \right) dv_{\leq i} \\
& + \sum_{i=1}^{n-1} \sum_{i'=i+1}^n \int_{v_{\leq i'}} \frac{H_{i'}}{b_{[i'] \setminus \{i\}}} \cdot g_i(v_{\leq i'}) \cdot \frac{2v_i}{b_i} dv_{\leq i'} \\
\text{s.t. } \quad & \frac{H_i}{h_i^0} \cdot Z_{i,i} \geq \Theta_i + \sum_{i'=i+1}^n \frac{H_{i'}}{h_i^{+1}} \cdot Z_{i,i'}, & \forall i \leq n \\
& Z_{i,i'} = \frac{1}{b_{[i'] \setminus \{i\}}} \cdot \int_{v_{\leq i'}} g_i(v_{\leq i'}) \, dv_{\leq i'}, & \forall i \leq i' \in [n]
\end{aligned}$$

$$g_i(v_{\leq i'}) = \mathbb{1}\left(i = \arg \max_{0 \leq i'' \leq i'} \left(\frac{v_{i''}}{b_{i''}} + \Delta_{i'', i'}\right)\right), \quad \forall i \leq i' \in [n], v_{\leq i'}.$$

We now solve this program using induction on n . For the base case, we have $n = 1$ and the program simplifies to:

$$\begin{aligned} \max \quad & \int_{v_1} g_1(v_1) \cdot \left(\frac{2v_1}{b_1} - 1 - \Phi_1\right) dv_1 \\ \text{s.t.} \quad & Z_{1,1} \geq \Theta_1, \\ & Z_{1,1} = \int_{v_1} g_1(v_1) dv_1, \\ & g_1(v_1) = \mathbb{1}\left(\frac{v_1}{b_1} + \Delta_{1,1} \geq 0\right), \quad \forall v_1. \end{aligned}$$

As $v_1 \in [0, b_1]$, we can assume without loss of generality that $\Delta_{1,1} \in [-1, 0]$. This gives:

$$\begin{aligned} \max \quad & b_1 \cdot (-\Phi_1 - \Delta_{1,1} \cdot (1 + \Phi_1) - \Delta_{1,1}^2) \\ \text{s.t.} \quad & b_1 \cdot (1 + \Delta_{1,1}) \geq \Theta_1, \\ & \Delta_{1,1} \in [-1, 0]. \end{aligned}$$

RRS: This is easily solvable and we use $\Upsilon_1(\Theta_1, \Phi_1)$ to denote the optimal value. Note that the function $\Upsilon_1(\cdot)$ is concave in Θ_1 for any fixed value of Φ_1 . This finishes the base case. For the inductive case, we show how to solve the optimization problem for n assuming we have the solution for $n - 1$. Let $\Upsilon_{n-1}(\Theta_{<n}, \Phi_{<n})$ be the solution for $n - 1$ promised by the induction hypothesis. For this, note that the optimization problem for n is the same as:

$$\begin{aligned} \max \quad & \Upsilon_{n-1}\left(\left(\Theta_i + \frac{H_n}{h_i^{+1}} \cdot Z_{i,n}\right)_{i < n}, \Phi_{<n}\right) \\ & + \int_{v_{\leq n}} \frac{H_n}{b_{<n}} \cdot g_n(v_{\leq n}) \cdot \left(\frac{2v_n}{b_n} - 1 - \Phi_n\right) dv_{\leq n} + \sum_{i=1}^{n-1} \int_{v_{\leq n}} \frac{H_n}{b_{[n] \setminus \{i\}}} \cdot g_i(v_{\leq n}) \cdot \frac{2v_i}{b_i} dv_{\leq n} \\ \text{s.t.} \quad & \frac{H_n}{h_n^0} \cdot Z_{n,n} \geq \Theta_n, \\ & Z_{i,n} = \frac{1}{b_{[n] \setminus \{i\}}} \cdot \int_{v_{\leq n}} g_i(v_{\leq n}) dv_{\leq n}, \quad \forall i \in [n] \\ & g_i(v_{\leq n}) = \mathbb{1}\left(i = \arg \max_{0 \leq i'' \leq n} \left(\frac{v_{i''}}{b_{i''}} + \Delta_{i'', n}\right)\right), \quad \forall i \in [n], v_{\leq n} \end{aligned}$$

RRS: Finish using concavity

4.5 Lemmas

Lemma 4.1. *For any feasible solution $(\text{pay}_i(\cdot), \alpha_i(\cdot))_{i \in N}$ of Equation (9), there exists another feasible solution $(\text{pay}'_i(\cdot), \alpha'_i(\cdot))_{i \in N}$ of Equation (9) that does not have a worse objective value and is also feasible for Equation (8).*

Proof. □

Lemma 4.2. *For any feasible solution $(\text{pay}_i(\cdot), \alpha_i(\cdot))_{i \in N}$ of Equation (9), there exists another feasible solution $(\text{pay}'_i(\cdot), \alpha'_i(\cdot))_{i \in N}$ of Equation (9) that does not have a worse objective value and is also feasible for Equation (10).*

Proof. □

5 Line Graph with Two Buyers and a Seller

Consider $N = \{A, B\}$ connected over a line graph with s as shown in Figure 1. The agents' valuations v_A and v_B are distributed as f_A and f_B with supports $[0, b_A]$ and $[0, b_B]$ respectively. Since r_B can take only one value, we omit that in the argument of α_B . To

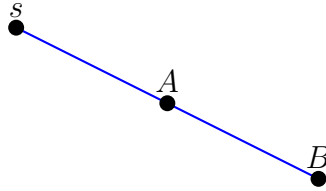


Figure 1: Line graph with two buyers and the seller

present our results more cleanly, we define a regularity condition.

Definition 5.1. *A valuation v is said to be regular if its distribution F (density represented by f) is such that*

1. $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ is monotone non-decreasing in v , and
2. $v(1 - F(v))$ is concave in v .

SN: we don't need both for all claims, but is it cleaner to use one consolidated definition rather than multiple ones? also, we can mention at the end of the proof that both conditions are not necessary for this result to hold. done!

In this simple setting, we can simplify ?? (using Lemma 3.5) to obtain

SN!

$$\begin{aligned}
\max \quad & h_A(\{B\})\text{pay}_A(\{B\}) + \sum_{i \in N} \sum_{r_i} \int_0^{b_i} h_i(r_i) f_i(v_i) \alpha_i(v_i, r_i) \varphi_i(v_i) dv_i \\
\text{s.t.} \quad & -\text{pay}_A(\{B\}) \geq \int_0^v (\alpha_A(y, \emptyset) - \alpha_A(y, \{B\})) dy, \quad \forall v \in [0, b_A], \\
& \text{pay}_A(\{B\}) \leq 0, \text{ and } (\alpha_i)_{i \in N} \in \mathcal{R},
\end{aligned} \tag{11}$$

5.1 Structure of the Allocation

In this section, our goal is to simplify the constraints in [Equation \(11\)](#). We show in [Theorem 5.2](#) firstly that we can obtain an optimal DA so that $\alpha_A(\cdot, \emptyset)$ is a step function. The expression on the right of the constraint in [Equation \(11\)](#) thus first decreases until the threshold for the step function, and increases afterwards, and is consequently maximum at either $v = 0$ or $v = b_A$. Secondly, the theorem shows that in fact the maximum occurs at $v = b_A$.

Theorem 5.2. *For every feasible solution $(\alpha_i, \text{pay}_i)_{i \in N}$ of ??, there is another feasible solution $((\alpha'_i)_{i \in N}, (\text{pay}'_i)_{i \in N})$ that does no worse in the value of the objective function, with*

1. *the function $\alpha'_A(\cdot, \emptyset)$ is a step function, and*
2. *the following inequality holds*

$$\int_0^{b_A} \alpha'_A(y, \emptyset) dy \geq \int_0^{b_A} \alpha'_A(y, \{B\}) dy.$$

In the rest of the paper, we will refer to the point x^* of a step function $T(x)$ where it switches from zero to unity, i.e., $T(x) = \mathbb{1}\{x \geq x^*\}$, as the *switchover point* of T .

Proof. Consider an arbitrary feasible solution $(\alpha_i, \text{pay}_i)_{i \in N}$ of ??. Define the step function $\alpha'(y)$ to satisfy:

$$\int_0^{b_A} \alpha'(y) dy = \int_0^{b_A} \alpha_A(y, \emptyset) dy.$$

Note that this indeed defines α' uniquely for any given $\alpha_A(y, \emptyset)$. To prove the claim, we show that replacing the function $\alpha_A(y, \emptyset)$ in our feasible solution by $\alpha'(y)$ (keeping everything else same as in the α function, i.e., $\alpha'_A(y, \{B\}) = \alpha_A(y, \{B\}), \forall y \in [0, b_A]$) maintains feasibility and can only increase the objective function. For this, we use the fact that our definition implies that for all $v_A \in [0, b_A]$, we have $\int_0^{v_A} \alpha'(y) dy \leq \int_0^{v_A} \alpha_A(y, \emptyset) dy$. Note that the only non-trivial constraint of ?? for agent A is where $r'_A = \emptyset$ and $r_A = \{B\}$. Hence, replacement of any feasible $\alpha_A(y, \emptyset)$ with the corresponding $\alpha'(y)$ as defined above will not violate the constraints. However, from [Fact 2.2](#), we know that

$$\int_0^{b_A} f_A(v_A) \varphi_A(v_A) \alpha'(v_A) dv_A \geq \int_0^{b_A} f_A(v_A) \varphi_A(v_A) \alpha_A(v_A, \emptyset) dv_A,$$

since, $f_A(v_A)\varphi_A(v_A)$ is monotone non-decreasing as $v_A(1 - F_A(v_A))$ is concave (condition 2 of Definition 5.1), and $\exists v_A^* \in [0, b_A]$ where $\alpha'(\cdot)$ and $\alpha_A(\cdot, \emptyset)$ crosses. This completes condition 1 of the proof.

To prove condition 2, consider the same arbitrary feasible solution $(\alpha_i, \text{pay}_i)_{i \in N}$ of ??, with $\alpha_A(\cdot, \emptyset)$ being replaced by the step function $\alpha'_A(\cdot, \emptyset)$ as explained in the proof of condition 1 above. Let y_1 be any point where φ_A crosses the zero point ($y_1 = 0$ if $\varphi_A(v_A) > 0, \forall v_A \in [0, b_A]$ and $y_1 = b_A$ if $\varphi_A(v_A) < 0, \forall v_A \in [0, b_A]$). Such a y_1 must exist if v_A is regular.

The proof of this part is constructive. Suppose, condition 2 does not hold for the choice of feasible solution after the above replacement, i.e.,

$$\int_0^{b_A} \alpha'_A(y, \emptyset) dy < \int_0^{b_A} \alpha_A(y, \{B\}) dy.$$

Then we construct another set of a feasible solutions that satisfy conditions 1 and 2, yet does not decrease the value of the objective function. In other words, if condition 2 fails for some step function received from condition 1, then there exists another step function that passes the statement of the theorem. Consider the following two exhaustive cases.

- **Case 1:** $b_A - y_1 \geq \int_0^{b_A} \alpha_A(y, \{B\}) dy$. Define the step function $\beta'(y)$ to satisfy:

$$\int_0^{b_A} \beta'(y) dy = \int_0^{b_A} \alpha_A(y, \{B\}) dy.$$

Note that such a step function β' is unique (having a switchover point at $y_2 \geq y_1$), and we claim that substituting $\beta'(y)$ instead of $\alpha'_A(y, \emptyset)$ in the optimization problem of ?? can only improve the objective function and maintains all feasibility constraints. To see this, consider the constraint of agent A :

$$\text{pay}_A(\emptyset) - \text{pay}_A(\{B\}) \geq \int_0^{v_A} (\alpha'_A(y, \emptyset) - \alpha_A(y, \{B\})) dy, \quad (12)$$

which should hold for all v_A . In particular, it should hold at that v_A where the integral on the RHS is maximum. However, note that since $\alpha'_A(y, \emptyset)$ is a step function and both $\alpha'_A(y, \emptyset)$ and $\alpha_A(y, \{B\})$ are monotone non-decreasing, the RHS can be either maximized at $v_A = 0$ or at $v_A = b_A$. But we already know that at $v_A = b_A$, the RHS is negative. Hence, the maximum value of zero is achieved at $v_A = 0$. When we substitute $\alpha'_A(y, \emptyset)$ with $\beta'(y)$, we do not change the RHS, and hence, the solution remains feasible. However, since $\beta'(\cdot)$ is also a step function having a larger integral $\int_0^{b_A} \beta'(y) dy$ than $\int_0^{b_A} \alpha'_A(y, \emptyset) dy$, its switchover point y_2 is smaller than the switchover point of $\alpha'_A(y, \emptyset)$. Therefore, the substitution weakly improves the objective function of ?? as $\varphi_A(y)$ is non-negative for $y \geq y_2 \geq y_1$.

- **Case 2:** $b_A - y_1 < \int_0^{b_A} \alpha_A(y, \{B\}) dy$. In this case, define the value y_2 to be such that:

$$b_A - y_1 = \int_{y_2}^{b_A} \alpha_A(y, \{B\}) dy.$$

Note that this indeed defines y_2 uniquely and that we must have $y_2 \leq y_1$. Define the functions $\beta'_\emptyset(y)$ and $\beta'_{\{B\}}(y)$ to be such that:

$$\begin{aligned} \beta'_\emptyset(y) &= \mathbb{1}(y \geq y_1). \\ \beta'_{\{B\}}(y) &= \begin{cases} 0, & \text{if } y \leq y_2 \\ \alpha_A(y, \{B\}), & \text{if } y > y_2 \end{cases}. \end{aligned}$$

Observe that using $\beta'_\emptyset$ instead of $\alpha'_A(y, \emptyset)$ and $\beta'_{\{B\}}$ instead of $\alpha_A(y, \{B\})$ can only improve the objective function and maintains all feasibility constraints. This follows using a similar argument as case 1. As argued there, the RHS of Equation (12) is maximized at $v_A = 0$, which makes the RHS equal to zero. The construction above keeps the RHS set to zero for case 2. However, since the newly constructed α_A functions only removes probabilities at or below y_1 from where the virtual valuation φ_A starts becoming positive, it can only improve the objective function of ???. This completes the proof of condition 2.

Together with conditions 1 and 2, we have the theorem. □

Given the above theorem, the following corollary is immediate.

Corollary 5.3. *There exists an optimal solution $(\alpha_i, \text{pay}_i)_{i \in N}$, such that:*

1. *the function $\alpha_A(\cdot, \emptyset)$ is a step function, and*
2. *the following inequality holds*

$$\int_0^{b_A} \alpha_A(y, \emptyset) dy \geq \int_0^{b_A} \alpha_A(y, \{B\}) dy.$$

For the optimal expected allocation functions $\alpha_A(\cdot, \emptyset)$, $\alpha_A(\cdot, \{B\})$ as in Corollary 5.3, let the switchover point of $\alpha_A(\cdot, \emptyset)$ be $\bar{v}_A$. Then $\int_0^{b_A} \alpha_A(y, \emptyset) dy = b_A - \bar{v}_A$. Let $\bar{t} = b_A - \bar{v}_A - \int_0^{b_A} \alpha_A(y, \{B\}) dy$, and note that $\bar{t} \geq 0$. In the next section, given $\bar{v}_A$ and $\bar{t}$, we will show how to obtain the optimal DA (g, p) .

SN: done this far

SN!

5.2 Obtaining (g, p) given $\bar{v}_A, \bar{t}$

We first simplify Equation (11) using Corollary 5.3 to obtain

$$\begin{aligned} \max \quad & h_A(\{B\})\text{pay}_A(\{B\}) + \sum_{i \in N} \sum_{r_i} \int_0^{b_i} h_i(r_i) f_i(v_i) \alpha_i(v_i, r_i) \varphi_i(v_i) dv_i \\ \text{s.t.} \quad & -\text{pay}_A(\{B\}) \geq \int_0^v (\alpha_A(y, \emptyset) - \alpha_A(y, \{B\})) dy, \quad \forall v \in [0, b_A], \\ & \text{pay}_A(\{B\}) \leq 0, \text{ and } (\alpha_i)_{i \in N} \in \mathcal{R}, \end{aligned}$$

The objective is maximized when $\text{pay}_A(\{B\})$ is maximized, and hence the upper bound on $\text{pay}_A(\{B\})$ must be tight. By definition of $\bar{t}$, this is equivalent to

$$\begin{aligned} \max \quad & h_A(\{B\})\bar{t} + \sum_{i \in N} \sum_{r_i} \int_0^{b_i} h_i(r_i) f_i(v_i) \alpha_i(v_i, r_i) \varphi_i(v_i) dv_i \\ \text{s.t.} \quad & \int_0^{b_A} \alpha_A(y, \{B\}) dy = b_A - \bar{v}_a - \bar{t} \\ & \text{and } (\alpha_i)_{i \in N} \in \mathcal{R}. \end{aligned}$$

Since $h_A(\{B\})$ and $\bar{t}$ are fixed, we can drop the first term from our optimization objective. Further, we can substitute $\alpha_A(y, \emptyset) = \mathbb{1}(y \geq \bar{v}_A)$ to obtain

$$\begin{aligned} \max \quad & \int_{\bar{v}_A}^{b_A} h_A(\emptyset) f_A(v_A) \varphi_A(v_A) dv_A + \int_0^{b_B} f_B(v_B) \alpha_B(v_B, \emptyset) \varphi_B(v_B) dv_B \\ & + \int_0^{b_A} h_A(\{B\}) f_A(v_A) \alpha_A(v_A, \{B\}) \varphi_A(v_A) dv_A \\ \text{s.t.} \quad & \int_0^{b_A} \alpha_A(y, \{B\}) dy = b_A - \bar{v}_a - \bar{t} \\ & (\alpha_i)_{i \in N} \in \mathcal{R}. \end{aligned}$$

From Equation (3), we can substitute the expected allocation functions α_A and α_B to obtain:

$$\begin{aligned} \max \quad & \int_{\bar{v}_A}^{b_A} h_A(\emptyset) f_A(v_A) \varphi_A(v_A) dv_A \\ & + \int_0^{b_A} \int_0^{b_B} h_A(\{B\}) f_A(v_A) f_B(v_B) g_B((v_A, \{B\}), (v_B, \emptyset)) \varphi_B(v_B) dv_B dv_A \\ & + \int_0^{b_A} \int_0^{b_B} h_A(\{B\}) f_A(v_A) f_B(v_B) g_A((v_A, \{B\}), (v_B, \emptyset)) \varphi_A(v_A) dv_B dv_A \\ \text{s.t.} \quad & \int_0^{b_A} \int_0^{b_B} f_B(v_B) g_A((v_A, \{B\}), (v_B, \emptyset)) dv_B dv_A = b_A - \bar{v}_A - \bar{t}, \end{aligned}$$

$$g_A((v_A, \{B\}), (v_B, \emptyset)) + g_B((v_A, \{B\}), (v_B, \emptyset)) \leq 1 \quad \forall v_A, v_B,$$

$$0 \leq g_A, g_B \leq 1$$

In order to identify optimal allocation functions, define the function $Z(v_A, v_B) = f_A(v_A)(\varphi_A(v_A) - \max(0, \varphi_B(v_B)))$. We define the allocation function g_A, g_B as follows:

$$g_A((v_A, \{B\}), (v_B, \emptyset)) = \mathbf{1}(Z(v_A, v_B) \geq \bar{z})$$

$$g_B((v_A, \{B\}), (v_B, \emptyset)) = \mathbf{1}(Z(v_A, v_B) < \bar{z}) \cdot \mathbf{1}(\varphi_B(v_B) \geq 0)$$

where $\bar{z}$ is defined so that the constraint in the optimization problem is satisfied, i.e.,

$$\int_0^{b_A} \int_0^{b_B} f_B(v_B) \mathbf{1}(Z(v_A, v_B) \geq \bar{z}) dv_B dv_A = b_A - \bar{v}_A - \bar{t}.$$

Recall that $g_A((v_A, \emptyset), (v_B, \emptyset)) = \mathbf{1}(v_A \geq \bar{v}_A)$. Clearly, this allocation function satisfies the optimization constraints. **UB:** Need to show that $\bar{z}$ exists, and that g is monotone, etc. **UB!**

We claim that the allocation function defined is actually optimal, given $\bar{v}_A$ and $\bar{t}$. To see this, assume for a contradiction that there exists a function g' that strictly improves on the objective. Then

$$\begin{aligned} & \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) g_B((v_A, \{B\}), (v_B, \emptyset)) \varphi_B(v_B) dv_B dv_A \\ & + \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) g_A((v_A, \{B\}), (v_B, \emptyset)) \varphi_A(v_A) dv_B dv_A \\ & < \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) g'_B((v_A, \{B\}), (v_B, \emptyset)) \varphi_B(v_B) dv_B dv_A \\ & \quad + \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) g'_A((v_A, \{B\}), (v_B, \emptyset)) \varphi_A(v_A) dv_B dv_A \end{aligned}$$

As $g_B((v_A, \{B\}), (v_B, \emptyset)) \varphi_B(v_B) = (1 - g_A((v_A, \{B\}), (v_B, \emptyset))) \cdot \max(0, \varphi_B(v_B))$ by definition and we have the constraint $g'_B((v_A, \{B\}), (v_B, \emptyset)) \leq 1 - g'_A((v_A, \{B\}), (v_B, \emptyset))$, we have that:

$$\begin{aligned} & \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) (1 - g_A((v_A, \{B\}), (v_B, \emptyset))) \cdot \max(0, \varphi_B(v_B)) dv_B dv_A \\ & + \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) g_A((v_A, \{B\}), (v_B, \emptyset)) \varphi_A(v_A) dv_B dv_A \\ & < \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) (1 - g'_A((v_A, \{B\}), (v_B, \emptyset))) \max(0, \varphi_B(v_B)) dv_B dv_A \\ & \quad + \int_0^{b_A} \int_0^{b_B} f_A(v_A) f_B(v_B) g'_A((v_A, \{B\}), (v_B, \emptyset)) \varphi_A(v_A) dv_B dv_A \end{aligned}$$

Simplifying the above inequality, we get

$$\begin{aligned} & \int_0^{b_A} \int_0^{b_B} f_B(v_B) g_A((v_A, \{B\}), (v_B, \emptyset)) f_A(v_A) (\phi_A(v_A) - \max(0, \varphi_B(v_B))) \, dv_B \, dv_A \\ & < \int_0^{b_A} \int_0^{b_B} f_B(v_B) g'_A((v_A, \{B\}), (v_B, \emptyset)) f_A(v_A) (\phi_A(v_A) - \max(0, \varphi_B(v_B))) \, dv_B \, dv_A \end{aligned}$$

Using the definition of $Z(v_A, v_B)$ and simplifying, we get:

$$\begin{aligned} & \int_0^{b_A} \int_0^{b_B} f_B(v_B) g_A((v_A, \{B\}), (v_B, \emptyset)) \cdot Z(v_A, v_B) \, dv_B \, dv_A \\ & < \int_0^{b_A} \int_0^{b_B} f_B(v_B) g'_A((v_A, \{B\}), (v_B, \emptyset)) \cdot Z(v_A, v_B) \, dv_B \, dv_A \end{aligned}$$

This is a contradiction because of the following lemma

Lemma 5.4 (Generalization of [Fact 2.2](#)). *Let $a_1 \leq b_1$ and $a_2 \leq b_2$ be given. Let $q, w, \lambda, \gamma : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be functions such that w is non-negative and:*

$$\int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \lambda(x, y) \, dy \, dx = \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \mu(x, y) \, dy \, dx,$$

and for which there exists a $z^ \in \mathbb{R}$ such that, for all $a_1 \leq x \leq b_1$ and all $a_2 \leq y \leq b_2$, we have:*

$$q(x, y) < z^* \implies \mu(x, y) \leq \lambda(x, y) \quad \text{and} \quad q(x, y) > z^* \implies \mu(x, y) \geq \lambda(x, y).$$

Then, we have:

$$\int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) q(x, y) \lambda(x, y) \, dy \, dx \leq \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) q(x, y) \mu(x, y) \, dy \, dx.$$

To use the lemma, we set $w(x, y) = f_B(v_B)$, $\lambda(x, y) = g_A((v_A, \{B\}), (v_B, \emptyset))$, $\mu(x, y) = g'_A((v_A, \{B\}), (v_B, \emptyset))$, and $q(x, y) = Z(v_A, v_B)$, $z^* = \bar{z}$. Note that then, by the optimization constraints,

$$\int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \lambda(x, y) \, dy \, dx = \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \mu(x, y) \, dy \, dx = b_A - \bar{v}_A - \bar{t}.$$

Proof of Lemma 5.4. Note that:

$$\int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) q(x, y) \cdot (\lambda(x, y) - \mu(x, y)) \, dy \, dx$$

$$\begin{aligned}
&= \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) q(x, y) \cdot (\lambda(x, y) - \mu(x, y)) \cdot \mathbb{1}(q(x, y) < z^*) \, dy \, dx \\
&\quad + \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) q(x, y) \cdot (\lambda(x, y) - \mu(x, y)) \cdot \mathbb{1}(q(x, y) = z^*) \, dy \, dx \\
&\quad + \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) q(x, y) \cdot (\lambda(x, y) - \mu(x, y)) \cdot \mathbb{1}(q(x, y) > z^*) \, dy \, dx \\
&\leq \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \cdot z^* \cdot (\lambda(x, y) - \mu(x, y)) \cdot \mathbb{1}(q(x, y) < z^*) \, dy \, dx \\
&\quad + \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \cdot z^* \cdot (\lambda(x, y) - \mu(x, y)) \cdot \mathbb{1}(q(x, y) = z^*) \, dy \, dx \\
&\quad + \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \cdot z^* \cdot (\lambda(x, y) - \mu(x, y)) \cdot \mathbb{1}(q(x, y) > z^*) \, dy \, dx \\
&= \int_{a_1}^{b_1} \int_{a_2}^{b_2} w(x, y) \cdot z^* \cdot (\lambda(x, y) - \mu(x, y)) \, dy \, dx \\
&= 0.
\end{aligned}$$

□

Theorem 5.5. *For a given change-over point $\bar{v}_A$, the allocation structure of the item among the two agents is a linearly separated region.*

Proof. TODO

□

SN: We need an example for the uniform distribution for the region of optimal allocation. **SN!**
This is for illustration with a figure perhaps.

Discussion. Note that condition 2 of Definition 5.1 is not necessary for agent B for these results to hold. However, we kept the definition of regularity such that it applies to more general settings that we address next.